cBioPortal
FOR CANCER GENOMICS

Tutorial 7
Pathways

Explore genomic data in the context of pathways

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cbioportal.org/tutorials

Tutorial objectives

- Explain why it helps to look at genomic data in the context of pathways
- Introduce the two tools on the Pathways tab, PathwayMapper and NDEx Cancer Pathways: each has its own pathways, views and ranking
- Walk through each tool: its pathway view, its toolbar and how it ranks pathways
- Find the Pathways tab in the Results View and in the Patient View

Why pathways?

- Alterations in cancer tend to hit a small number of signaling pathways: cell proliferation, cell growth, apoptosis, DNA repair.
- Seeing your genes on a pathway shows which other genes in it are altered too, and which pathways your genes have in common.
- The Pathways tab draws your genes on pathways from two tools, with how often each gene is altered in your query (Results View) or whether it is altered in one patient (Patient View).

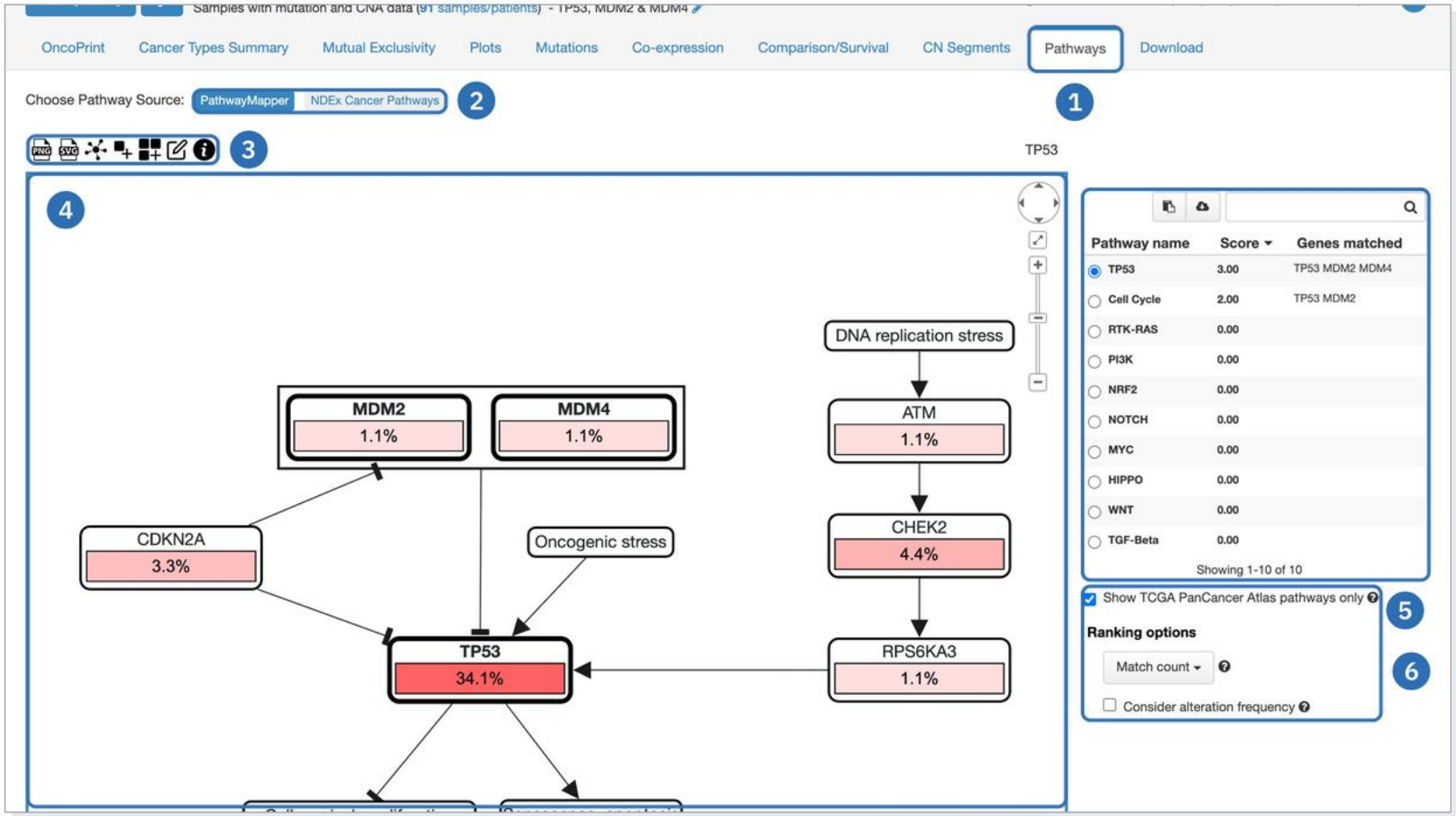
Two pathway tools

- PathwayMapper: pathway diagrams curated by The Cancer Genome Atlas (TCGA), the ten PanCancer Atlas pathways and those of other TCGA publications. Pathways are ranked by how well they match the query genes, and can be edited in the PathwayMapper editor (pathwaymapper.org).
- NDEx Cancer Pathways: networks from NDEx, the Network Data Exchange (ndexbio.org), collected from sources such as NCI-PID, Signor, WikiPathways, NeST and INDRA GO. Networks are ranked by similarity to the query genes, p-value or overlap, and can be opened in Cytoscape.
- Both show the query genes with their alteration frequency in the queried samples. The Patient View's Pathways tab uses PathwayMapper.

The Pathways tab in the Results View

- Example: TP53, MDM2 and MDM4 in Glioblastoma (TCGA, Nature 2008), samples with mutation and CNA data.
- Not sure how to run a query? See Tutorial 2.
- Click the Pathways tab. PathwayMapper opens first; NDEx Cancer Pathways is one click away.

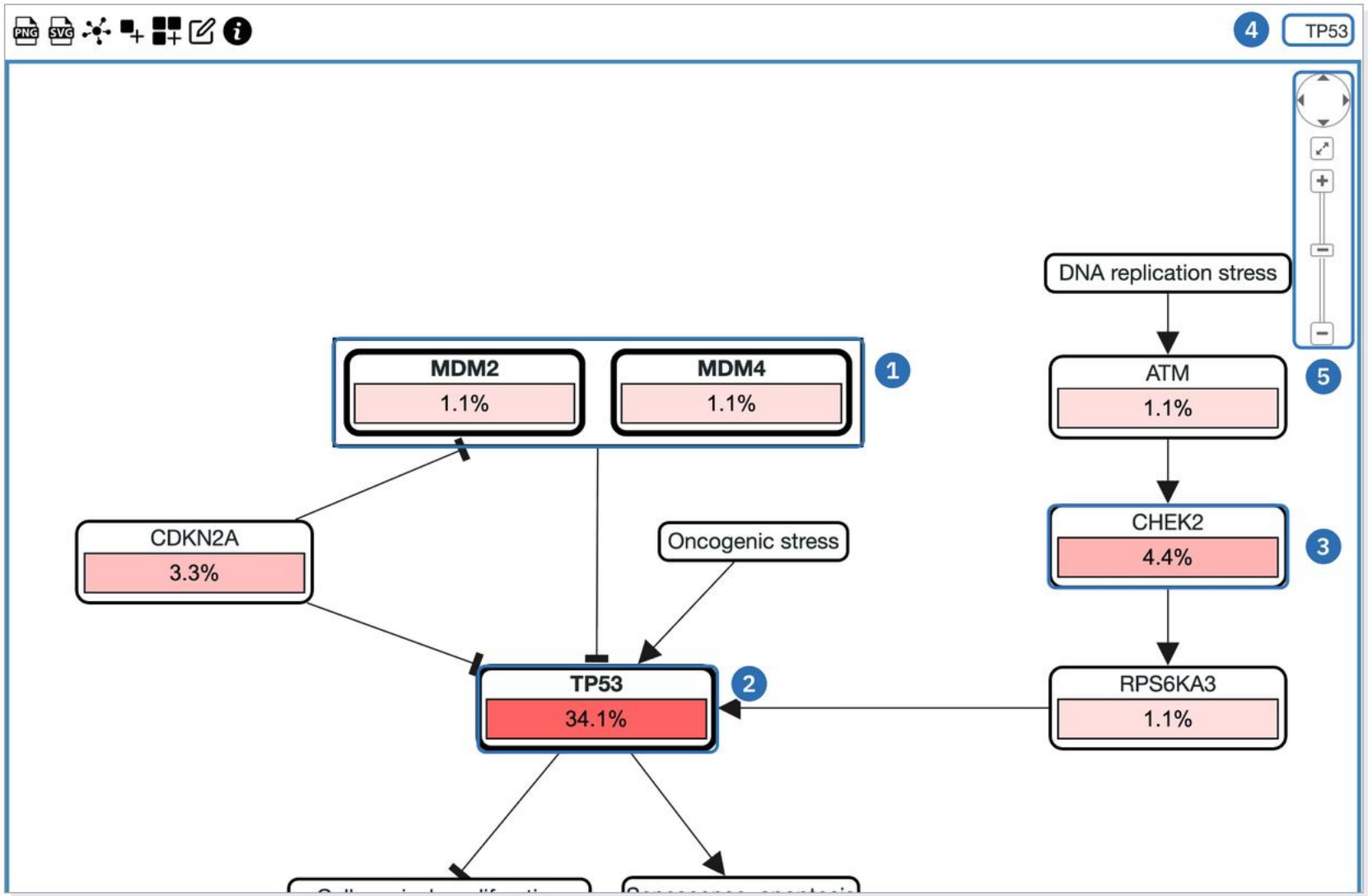
PathwayMapper: overview



1. The Pathways tab.
2. Choose the tool: “PathwayMapper” (the default) or “NDEx Cancer Pathways”.
3. The toolbar (next slides).
4. The pathway, with the alteration frequency of each gene in the queried samples.
5. The pathways, ranked by how well they match the query. Pick one to show it.
6. Which pathways to list, and how to rank them (next slides).

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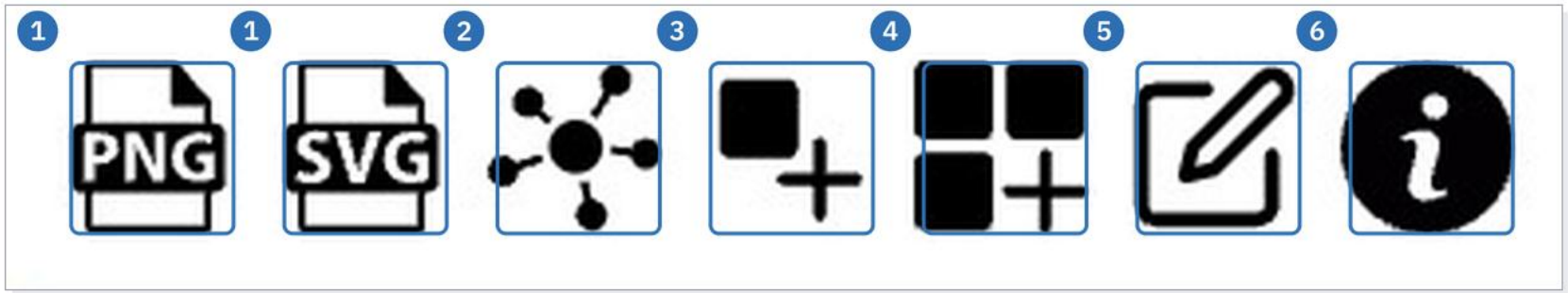
PathwayMapper: reading the pathway



1. Query genes have a thick border: MDM2 and MDM4 (here grouped, 1.1% each)...
2. ...and TP53, altered in 34.1% of samples.
3. Other genes in the pathway are shown with their own alteration frequency: the redder the box, the more often altered.
4. The name of the pathway shown: TP53.
5. Pan and zoom.

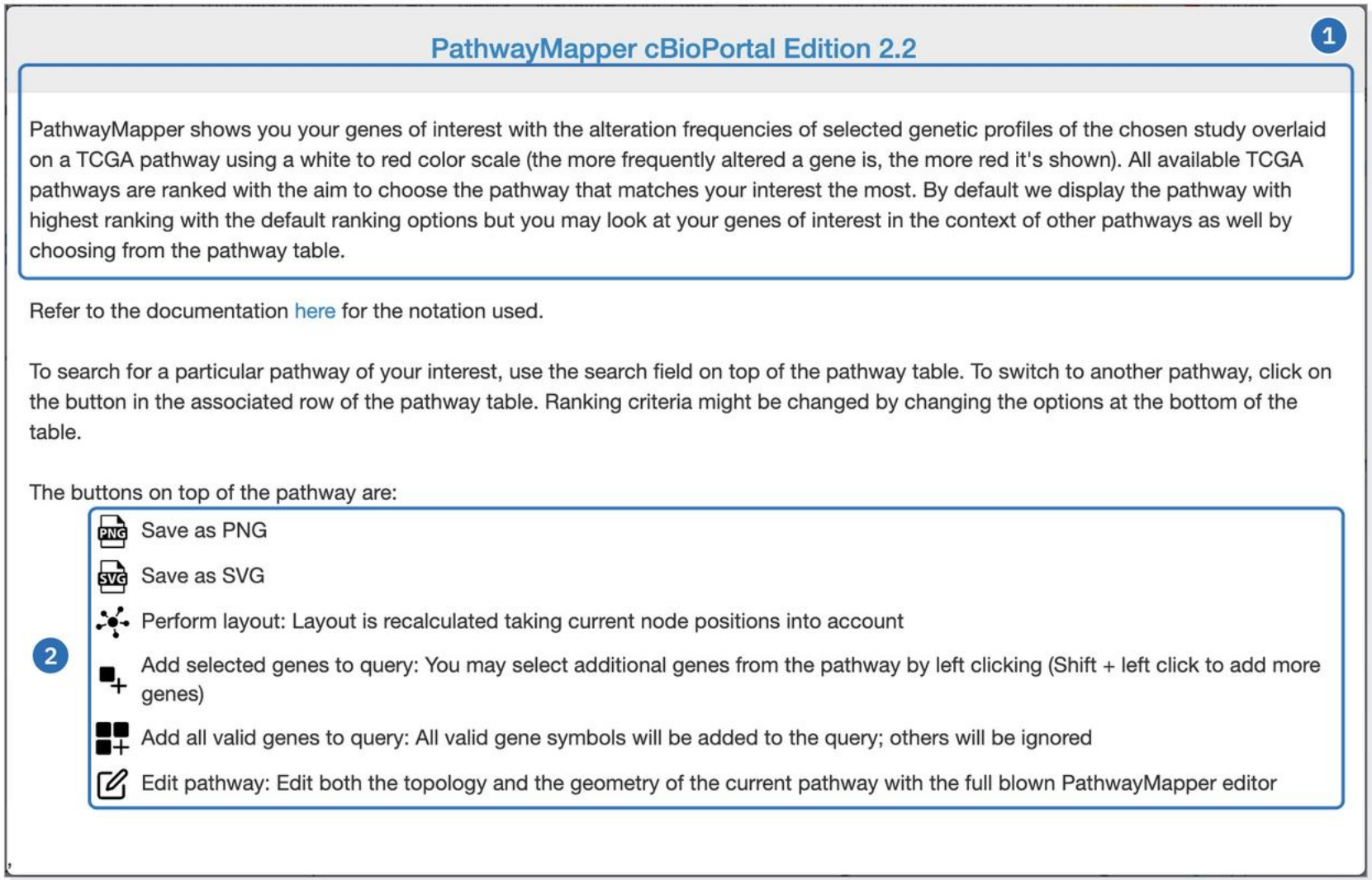
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PathwayMapper: the toolbar



1. Save the pathway as PNG or SVG.
2. Perform layout: recalculate the layout, keeping the current positions in mind.
3. Add selected genes to the query: click genes in the pathway (Shift-click for more), then rerun the query with them.
4. Add all valid genes in this pathway to the query.
5. Edit pathway: open it in the full PathwayMapper editor.
6. Help: what the colors and buttons mean, with a link to the notation.

PathwayMapper: help



PathwayMapper cBioPortal Edition 2.2

1

PathwayMapper shows you your genes of interest with the alteration frequencies of selected genetic profiles of the chosen study overlaid on a TCGA pathway using a white to red color scale (the more frequently altered a gene is, the more red it's shown). All available TCGA pathways are ranked with the aim to choose the pathway that matches your interest the most. By default we display the pathway with highest ranking with the default ranking options but you may look at your genes of interest in the context of other pathways as well by choosing from the pathway table.

Refer to the documentation here for the notation used.

To search for a particular pathway of your interest, use the search field on top of the pathway table. To switch to another pathway, click on the button in the associated row of the pathway table. Ranking criteria might be changed by changing the options at the bottom of the table.

The buttons on top of the pathway are:

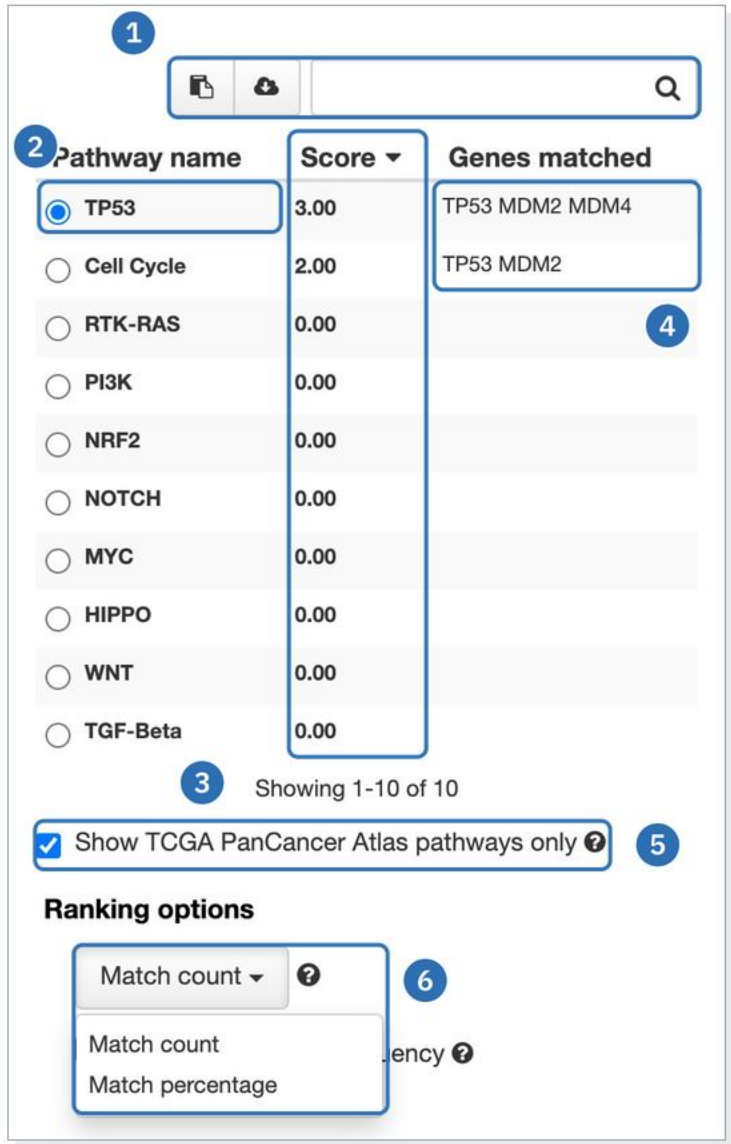
2

- Save as PNG
- Save as SVG
- Perform layout: Layout is recalculated taking current node positions into account
- Add selected genes to query: You may select additional genes from the pathway by left clicking (Shift + left click to add more genes)
- Add all valid genes to query: All valid gene symbols will be added to the query; others will be ignored
- Edit pathway: Edit both the topology and the geometry of the current pathway with the full blown PathwayMapper editor

1. What the view shows, how pathways are ranked, and a link to the notation (the meaning of the arrows and boxes).
2. What each toolbar button does.

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PathwayMapper: the pathway table



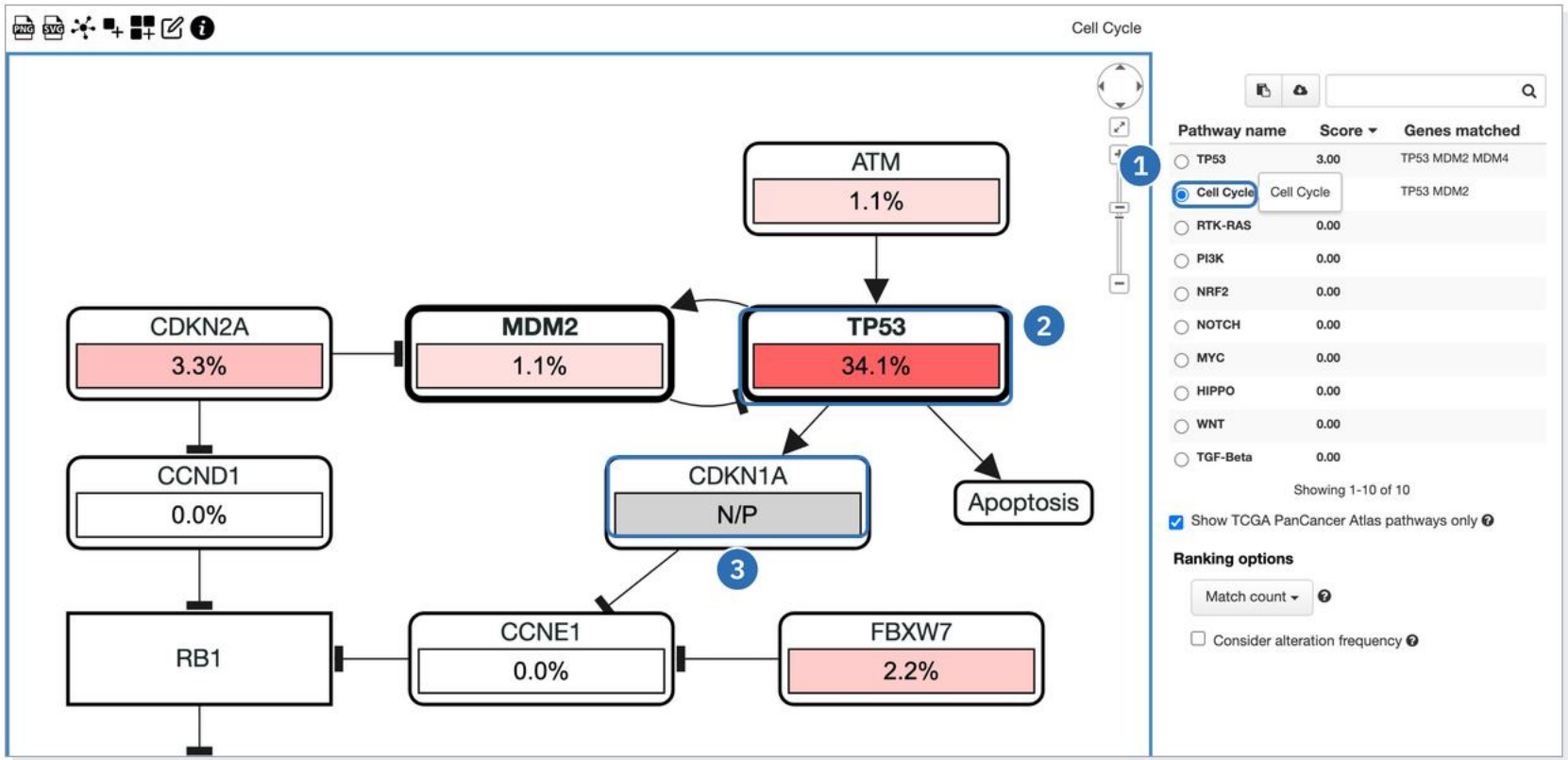
1. Copy or download the table, or search for a pathway by name.
2. The pathway shown (click another to switch).
3. Its score, with the current ranking options. Click the header to sort.
4. The query genes in each pathway.
5. Untick “Show TCGA PanCancer Atlas pathways only” to include the pathways from every TCGA publication.
6. Ranking options (next slide).

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PathwayMapper: ranking options

- A query gene that is in a pathway is a match. The TP53 pathway contains all three query genes, Cell Cycle two (TP53, MDM2).
- “Match count”: the number of matches. TP53 scores 3, Cell Cycle 2.
- “Match percentage”: matches as a share of the pathway's genes. TP53 scores 42.86 (3 of 7 genes), Cell Cycle 22.22.
- “Consider alteration frequency”: each match counts as its alteration frequency instead of 1. TP53 scores 36.27 (34.1 + 1.1 + 1.1), Cell Cycle 35.17.
- Q: Which pathway matches the query best?
  A: TP53, whichever option you use, so it is shown by default.

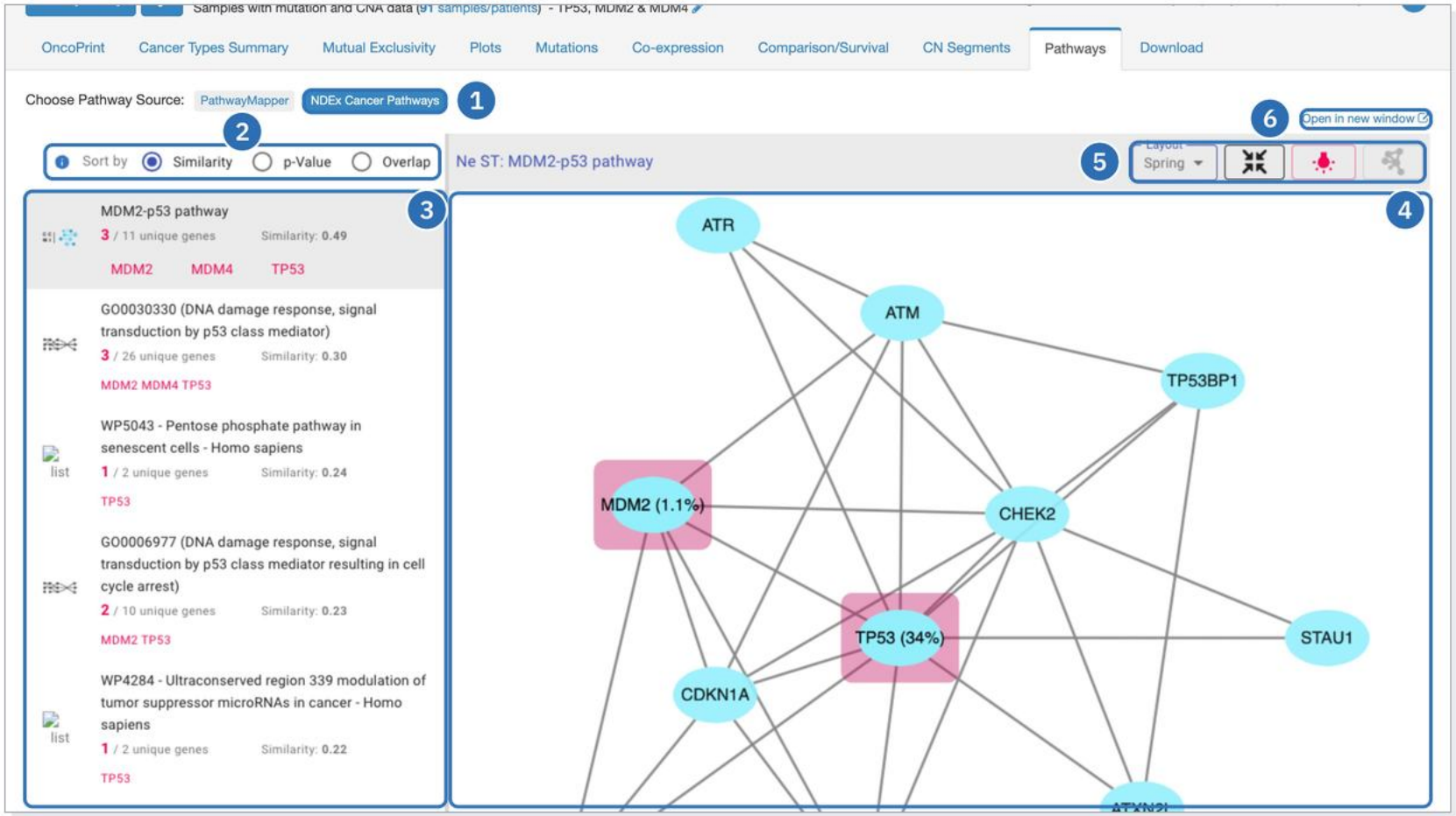
PathwayMapper: another pathway



1. Pick “Cell Cycle” in the table.
2. The same alteration frequencies on another pathway: here TP53 34.1% and MDM2 1.1% (query genes, thick borders), CDKN2A 3.3%, FBXW7 2.2%.
3. N/P: not profiled in this study.

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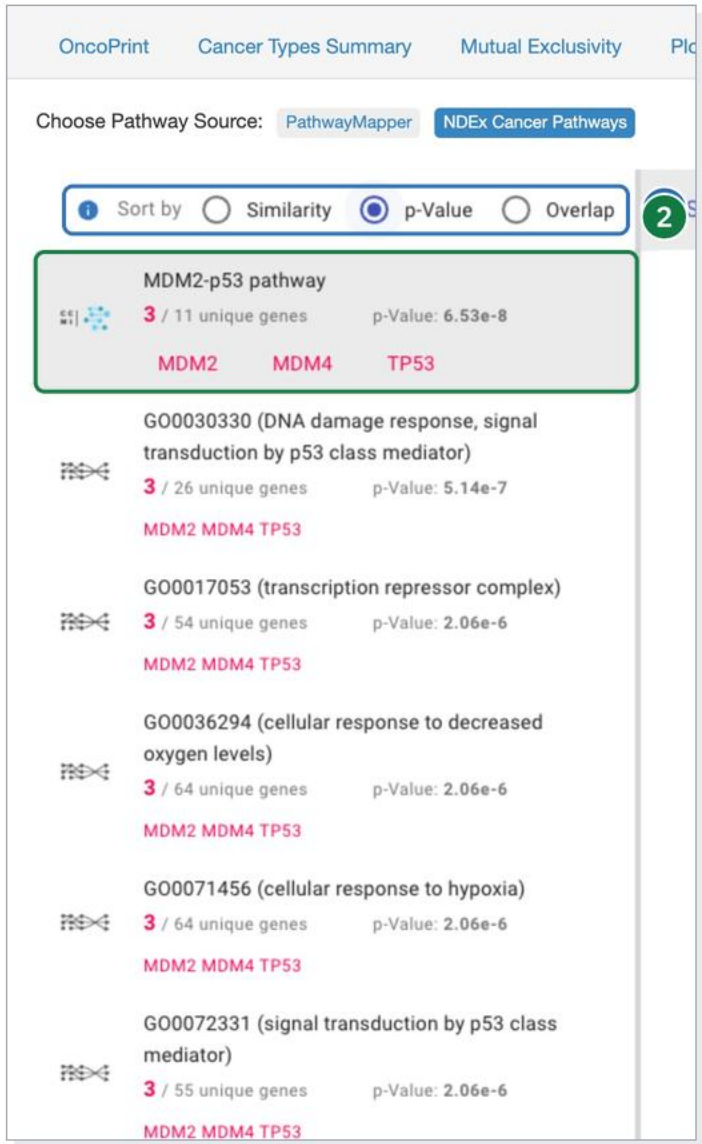
NDEx Cancer Pathways: overview



1. Choose the tool: “NDEx Cancer Pathways”.
2. How to rank the networks (next slides).
3. The networks that contain query genes, from NDEx's sources. Each shows how many query genes it has, out of its genes, and its score. Pick one to show it.
4. The network: query genes are pink boxes with their alteration frequency in the queried samples (TP53 34%, MDM2 1.1%).
5. The toolbar (next slides).
6. Open this view in a new window.

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NDEx Cancer Pathways: ranking

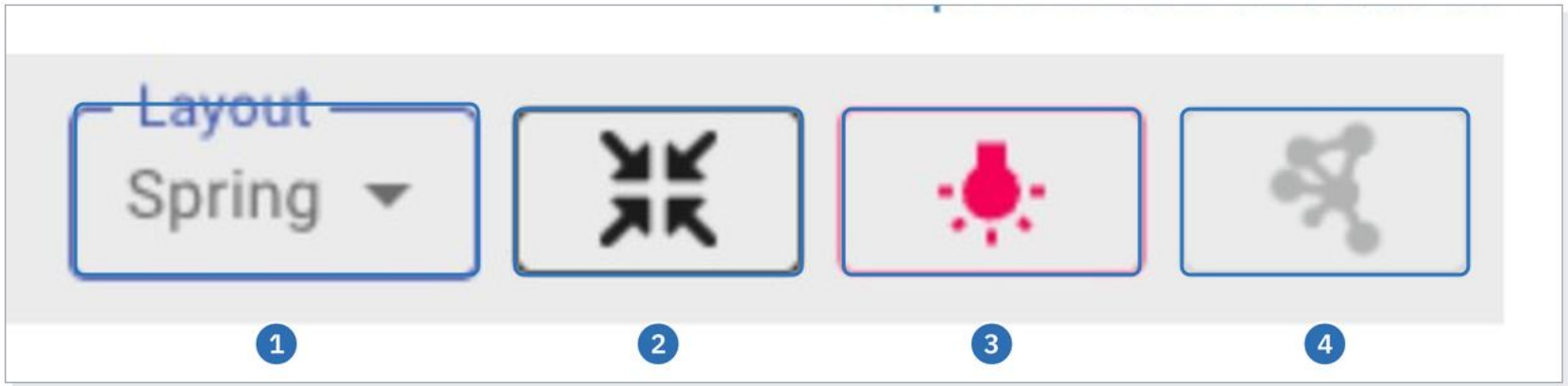


1. Sort by “Similarity” (how similar the network's genes are to the query genes; uncommon shared genes count more), “p-Value” (how unlikely the overlap is by chance, corrected for the many networks tested) or “Overlap” (how many query genes the network contains). Hover over the i icon for these definitions. The top 50 networks are listed.

2. Q: Which network matches the query best?
   A: The MDM2-p53 pathway: it contains all three query genes and comes first by similarity (0.49) and by p-value (6.53e-8).

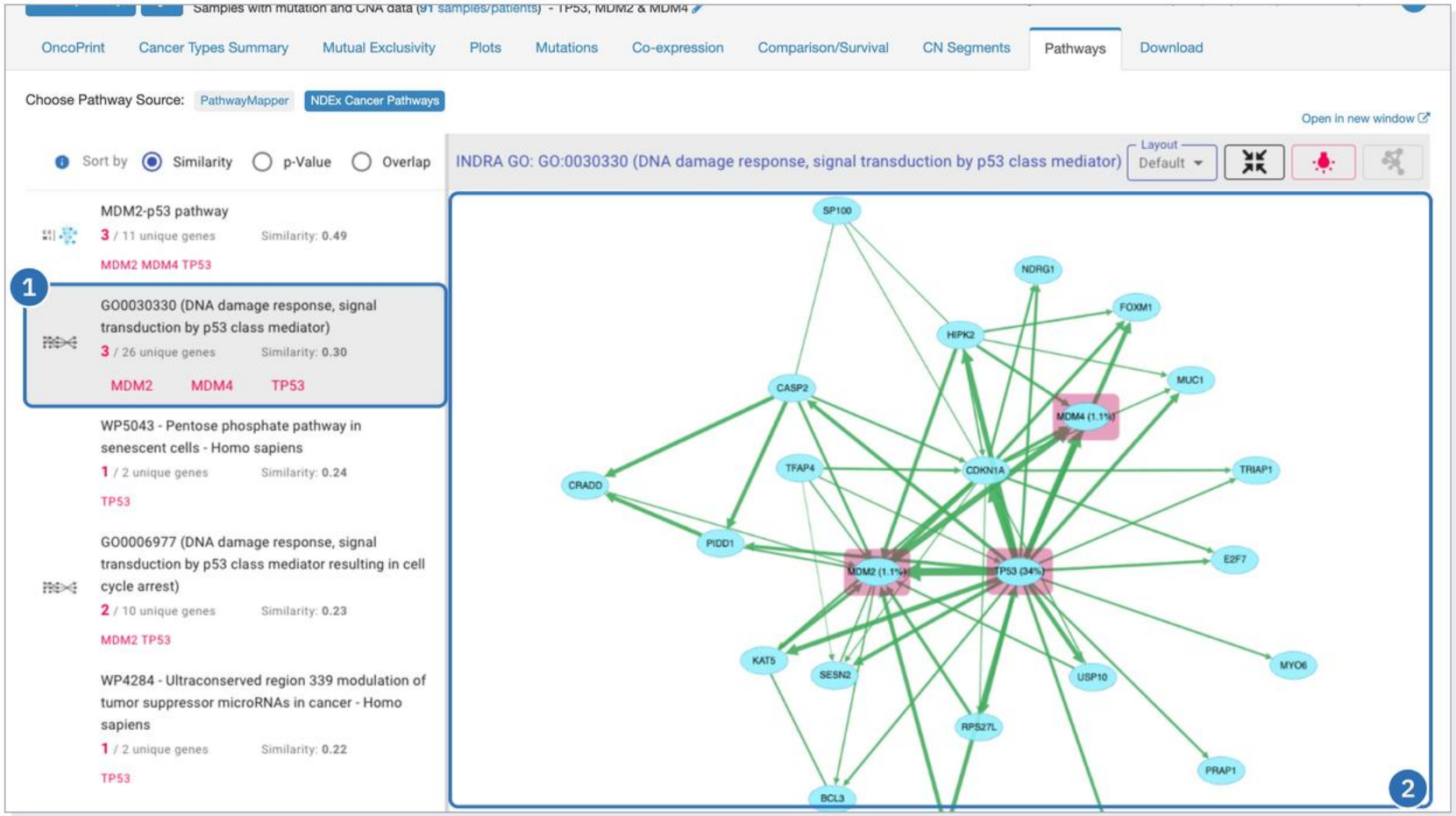
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NDEx Cancer Pathways: the toolbar



1. Layout: arrange the network as “Spring” or “Concentric”.
2. Fit the network to the panel.
3. Highlight the query genes.
4. Open the network in Cytoscape, the desktop app (needs Cytoscape 3.6 or later running, with the CyNDEx-2 app).

NDEx Cancer Pathways: another network



1. Pick another network: here GO0030330 (DNA damage response, signal transduction by p53 class mediator), from INDRA GO (its source is shown before its name, above the network).

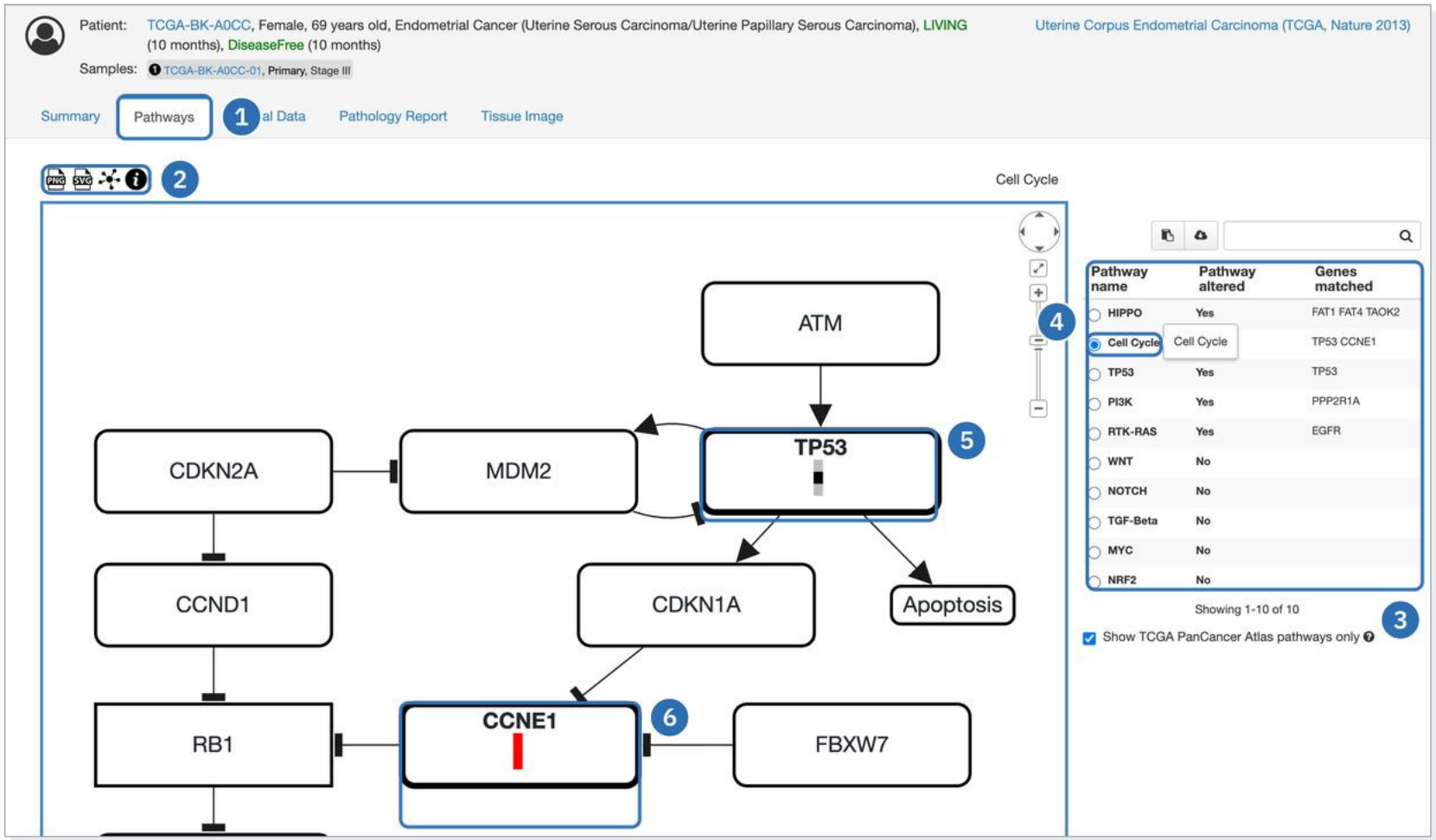
2. Each source draws its networks its own way: this one is larger, with arrows showing which gene acts on which.

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The Pathways tab in the Patient View

- The Patient View (Tutorial 3) has a Pathways tab too, with PathwayMapper's TCGA pathways. It marks the alterations of one patient.
- Example: TCGA-BK-A0CC, an endometrial cancer in Uterine Corpus Endometrial Carcinoma (TCGA, Nature 2013).
- Pathways where this patient’s tumor has alterations are listed first.

Patient View: the Pathways tab



1. The Pathways tab of the patient view.
2. Toolbar: save as PNG or SVG, layout, help.
3. The pathways, with whether this patient has an alteration in them: HIPPO, Cell Cycle, TP53, PI3K and RTK-RAS do.
4. Here: Cell Cycle.
5. Altered genes have a thick border and a mark for the alteration: TP53 is mutated (black)...
6. ...and CCNE1 is amplified (red). Hover over a gene for details.

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Questions?

PathwayMapper: pathwaymapper.org

NDEx: ndexbio.org

More tutorials: cbioportal.org/tutorials

Email us: cbioportal@googlegroups.com